Rational curiosity predicts the dynamics of habituation and dishabituation in infants and adults

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1 Abstract

How do we decide what to look at and when to stop looking? Even very young infants engage in active visual selection, looking less and less as stimuli are repeated (habituation) and regaining interest when novel stimuli are subsequently introduced (dishabituation). The mechanisms underlying these looking time changes remain uncertain, however, due to limits on both the scope of existing formal models and the empirical precision of measurements of infant behavior. We develop the Rational Action, Noisy Choice for Habituation (RANCH) model, which operates over raw images and makes quantitative predictions of participants' looking behaviors. In a series of pre-registered experiments, we measure infants' and adults' looking time across a range of stimulus exposure durations for both repeated and novel stimuli and show that they are predicted by RANCH. Our model also makes out-of-sample predictions about the magnitude of adult and infant dishabituation to a range of novel stimuli in an independent experiment. By framing looking behaviors as rational decision-making, this work identifies how the dynamics of learning and exploration guide our visual attention from infancy through adulthood.

2 Introduction

From birth, humans engage in active learning. Even before they gain independent mobility, infants make choices about what to look at and when to stop looking [20, 37]. Developmental psychologists have both studied this process of visual decision-making and also relied on it as a key methodological tool, particularly two key phenomena: habituation and dishabituation. Habituation occurs when infants show reduced interest upon repeated exposure to the same stimulus, whereas dishabituation entails renewed interest following the subsequent introduction of a novel stimulus. Dishabituation, also known as novelty preference, is often leveraged to infer infants' perceptual and cognitive abilities [2, 3, 16]: in order for an infant to dishabituate, they must distinguish between the original stimulus and the novel one. Despite the robust documentation of habituation and dishabituation in the literature, the mechanisms underlying these changes in gaze duration remain poorly understood. In this paper, we bridge this gap by introducing a rational model of habituation and dishabituation. We show that this rational model makes quantitative predictions of looking times toward different stimuli.

An important early conceptual model of infant looking time posited that habituation and dishabituation are influenced by the amount of information to be encoded by the infant from the stimulus [25]. Initially, infants look longer at stimuli containing substantial unprocessed information, showing a familiarity preference (Figure 1A). As exposure accumulates, less information remains unprocessed, leading to shorter looking time at the original stimulus and by comparison, relatively longer looking time to a new stimulus (i.e. novelty preference). While this model has had significant influence, the absence of specifications regarding the encoding process or a quantitative definition of information to be encoded have permitted post-hoc interpretations of looking time measurements. For instance, when infants are looking longer at stimulus A than stimulus B, researchers could interpret the looking time difference as either a novelty preference or a familiarity preference. This interpretive ambiguity has spurred repeated concerns regarding whether looking time measurements should form the basis for key assertions in developmental psychology [7, 21, 33]. In other words, developmental psychology needs formal models that make quantitative predictions, but the prevailing model proposed by Hunter and Ames [25] can not provide the necessary clarity to support robust inferences.

In contrast, more recent computational models aim to formalize the mechanisms underlying looking time changes. One family of models characterize infants' looking behaviors through information-theoretic metrics derived from ideal observer models: the models acquire probability distribution from event sequence and derive information-theoretic measures from the ideal learner's belief before and after each event [27, 34, 35]. For instance, Kidd, Piantadosi, and Aslin [27] developed a model that used surprise to quantify the extent to which the model learned from each observation. This measure correlated well with infants' looking behavior via a U-shaped curve: infants were least likely to look away when the surprisal of the event was neither too high nor too low (1B).

While these models provide quantitative fits to variation in looking time, they are not models of how an agent makes the decisions of whether to keep looking at the same stimulus or to look at something else. Instead, these models describe correlations between model-derived, information-theoretic measures and infant overall looking towards whole sequences of events. Relatedly, these models do not explain why infants would ever look at one static stimulus for a long time, rather than encode it instantaneously – there is no accounting of perceptual noise that needs to overcome by taking multiple perceptual samples that would justify why looking extends over time [c.f. 8, 26]. Owing to this constraint, these models can only be used to predict infant behavior in a specific paradigm: looking to a continuous stream of discrete stimuli. This paradigm is a significant deviation from the more common looking time paradigm in which the participants gradually encoded individual stimulus items in a self-paced manner. It is unclear whether the success of these quantitative models can be generalized to the more common looking time paradigms.

To address these issues, we developed the Rational Action, Noisy Choice for Habituation (RANCH) model [9]. RANCH describes an agent's looking behavior as rational exploration based on a sequence of noisy perceptual samples. The model construes the looking time paradigm as a series of binary decisions: to keep sampling from the current stimulus, or to look at anything else. The model makes sampling decisions based on the Expected Information Gain (EIG) of the upcoming perceptual sample, choosing to keep looking or look away based on which one would in expectation yield the most information; it therefore can be seen as a rational analysis of looking behavior [30, 31]. Rational analyses of behavior model decision-making as optimally adapted to the environment, given specific constraints. Identifying a behavior as rational provides a framework for predicting and influencing future actions, and facilitates the development of formal connections between human behavior and the environment in which it occurs [1].

Previously, we validated this rational analysis using a large dataset of adults engaging in a self-paced looking paradigm. We asked adults to look at sequences of stimuli consisting of a repeating familiar background stimulus (resulting in habituation), and a single, novel deviant stimulus after varying familiarization durations (resulting in dishabituation). Effects of prior exposure and novelty on adults' looking time were well captured by RANCH [9].

Here, we ask how well RANCH describes the dynamics of infant habituation and dishabituation, and compare it to other classic models of infant attention. Figure 1C shows schematic predictions of RANCH for the experimental setting described above: variable prior exposure to a familiar stimulus, followed by a novel or a familiar simulus. Rational exploration of a simple perceptual concept results in a monotonic decrease of attention to familiar items as a function of prior exposure.

One critical challenge for differentiating formal theories of looking time in infants is that looking time measurements are noisy, and many infant experiments provide only limited precision in their estimates of key quantities [10, 11]. In the current work, we develop a novel, within-subject experimental design that allows us to sensitively measure infants' looking time in experiments. Beyond validating RANCH's predictions in infants, this paradigm enables us to conduct interpretable developmental comparisons of parameter fits between infants and adults – since the experimental design was analogous to each other –

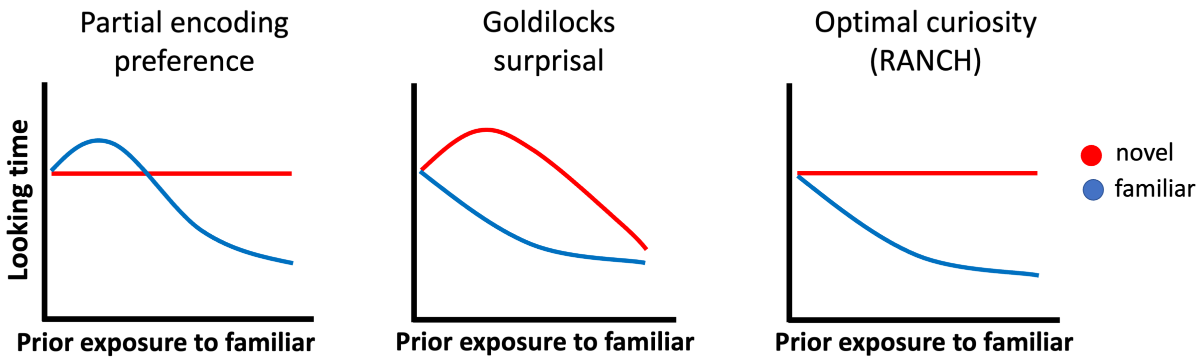

Figure 1: Conceptual models of looking to novel vs. familiar items according to three models of infant attention: A) The partial encoding model proposed by Hunter and Ames (1988) which suggests a shift from familiarity to novelty preferences, B) The Goldilocks model proposed by Kidd et al. (2012) which suggests that infants prefer to attend to intermediately surprising events, and C) the 'optimal curiosity' model introduced by Cao et al. (2023) which suggests that infants are maximizing expected information gain from noisy perceptual samples.

thereby characterizing how the mechanisms underlying looking time may change over development.

In addition, we incorporated recent progress in convolutional neural networks (CNNs) to allow RANCH to learn from raw images. Previous models have relied on using arbitrary representations of stimuli [e.g. object location, 27, hand-specified binary feature vector, 9]. However, recent CNNs have offered insights into how the visual system encodes objects [13, 23, 41]. The activations of these brain-inspired neural networks form embedding spaces, each of which can be seen as a quantitative hypothesis about how humans represent visual stimuli [38]. In RANCH, we leveraged these principled stimulus representations to enable RANCH to learn from raw images. In other words, RANCH can 'see' an experiment in a way similar to the one infants saw, and make predictions for previously unseen experiments using novel stimuli.

Last but not least, while previously we have shown that RANCH can successfully model habituation and dishabituation, we now test RANCH's ability to predict responses in new data. To conduct these tests, we first fit RANCH's parameters to a training data set from the habituation-dishabituation experiment in which participants saw sequences of stimuli. Then, we use the best-fitting parameters to generate predictions for a new experiment designed to measure subtle differences in dishabituation magnitude based on stimulus similarity. In this experiment, we systematically varied the similarity between habituation and dishabituation stimuli such that dishabituation stimuli differed in their pose angle, their number, their identity, or their animacy. This experiment tests a prediction from Hunter and Ames [25] model of habituation and dishabituation: that observers' dishabituation magnitude should be related to the similarity between the habituated stimulus and the novel stimulus. The more dissimilar two stimuli are, the more one should dishabituate to the novel stimulus.

The goal of this paper is to investigate the computational mechanisms underlying habituation and dishabituation by testing to what extent these behaviors can be explained by a rational analysis of looking time. To preview our results, we found support for the view that habituation and dishabituation are driven by rational curiosity by showing that RANCH predicts looking time in adults and infants. Furthermore, we also tested the importance of individual assumptions RANCH made via target ablation simulations. Removal of components of perceptual representation, learning progress, and decision making all result in substantially worse fit to data. Finally, we generate predictions for the novel task in which we vary the similarity between the habituation and dishabituation stimulus, and find that RANCH also captures the particular ordering of the dishabituation magnitude as a function of stimulus dissimilarity, thereby predicting a novel qualitative phenomenon (graded dishabituation) without ever being trained on it.

3 Rational Action, Nosy Choice for Habituation model

Rational Action, Noisy Choice for Habituation model (RANCH) is a Bayesian perception and action model in which a learner makes optimal perceptual sampling decisions [9]. In the current implementation, we conceptualized the learning problem in habituation as learning a visual concept – a probabilistic representation of the stimuli it experiences. The model achieves learning through observing a series of noisy perceptual samples from a sequence of stimuli. During the learning process, the model makes decisions about how many samples to receive of each stimulus before moving to the next stimulus. The number of samples that the model takes on each stimulus is then used as a proxy for looking time duration at each sample.

RANCH provides a probabilistic framework for making perceptual decisions. In this section, we will describe the three modular components RANCH includes: the perceptual representation, learning model, and decision model.

3.1 Perceptual representations

We use the deep convolutional neural network (CNN) ResNet-50 to encode the stimuli used in behavioral experiments. ResNet-50 is an architecture that has been widely used in computer vision [22]. A total of 50 layers are implemented in this architecture: it starts with a convolution layer, includes several groups of residual blocks that help preserve information through the network, and ends with a global average pooling layer and a fully connected layer for classification. We used a version of ResNet-50 that was pretrained on ImageNet [12]. Training on ImageNet optimizes the Resnet-50's activations for image classification, and we can use these activations as an perceptual "embedding space". When we pass new stimuli through a pre-trained network, we extract these activations to obtain principled embeddings of stimuli (Figure 2A). We used the first three principal components of the embedding space to limit the dimensionality of the downstream learning problem.

ResNet-50 is by no means the only possible CNN for deriving perceptual representations. We also explored alternative models, such as a CNN trained on SayCam data [32] or a CNN whose representations were aligned to human behavior [29]. However, we did not find substantial differences between the embedding spaces provided by ResNet-50 and other (often more complicated) models (Figure @ref(fig:embedding_distances)).

3.2 Learning model

In the current instantiation, RANCH models looking behaviors in habituation experiments as Bayesian concept learning [19, 39]. The model learns the concept through a series of noisy perceptual samples taken from a stimulus. The concept to be learned is parameterized by μ, σ, which represents beliefs about the location and variance of the presented concept in the embedding space. The concept μ, σ generates exemplars y, which is equivalent to the stimulus seen by the human participants. But RANCH does not directly learn from the stimulus. Instead, it learns from the series of noisy perceptual samples $\bar{z}$ generated by the exemplar. The noisy perceptual sample is corrupted by zero-mean noise, represented by ϵ. A plate diagram is shown in Figure 2B. At each time step, the model takes one perceptual sample to update the prior following Bayes Rule:

$$p(\mu, \sigma|\bar{z}) = \frac{p(\bar{z}|\mu, \sigma) * p(\mu, \sigma)}{p(\bar{z})}$$

3.3 Decision model and linking hypothesis

The Bayesian learning model did not address how the model decides to keep sampling the same stimulus or to move on to the next stimulus. We enable RANCH's decision-making process by calculating the expected information gain (EIG) of the next perceptual sample, which we define as the average KL-divergence across a grid of possible next observations. EIG is a metric used under the rational analysis of information-seeking behavior [1, 31]. RANCH then uses a softmax choice (with temperature = 1) between the information theoretic measures of the next sample and a constant "environmental EIG". The constant here is assumed to be the amount of information gain provided by the environment when the learner looks away from the stimulus. An algorithm table describing the overall sampling procedure is shown in (Figure 2C).

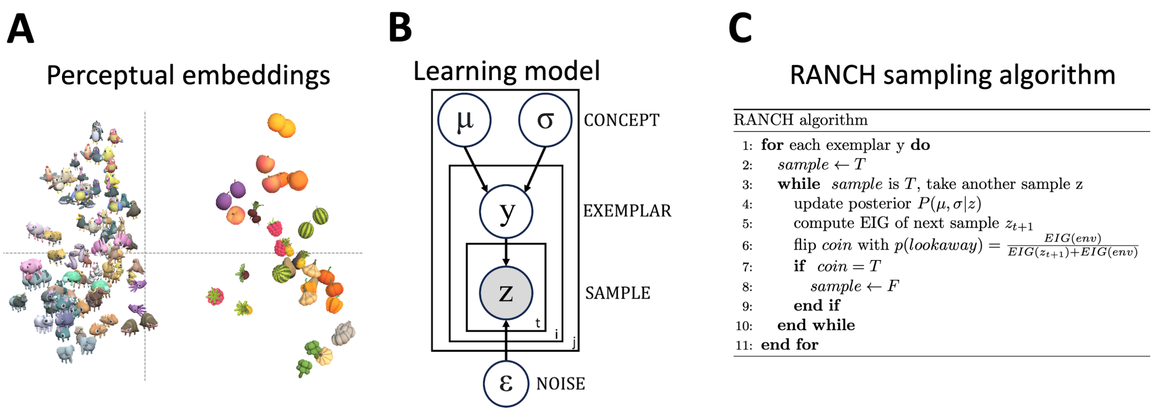

Figure 2: Three components of RANCH: A) Perceptual representation: Obtaining stimulus embeddings from ResNet-50 final layer activations, B) Learning model: Plate diagram for Gaussian concept learning from noisy observations, C) Decision model: RANCH repeatedly samples until environmental EIG outweighs EIG from another sample.

3.4 Implementation

RANCH has a series of parameters: the learner's priors over the concept to be learned (μ, σ), the learner's priors over their perceptual noise (σ_ϵ) and the environmental EIG (EIG_{env}). We conducted an iterative grid search over these parameters.

Given that the element of accounting for the learner's own noise breaks the conjugacy of updating the concept mean and variance, we resorted to grid approximation to perform the sample-by-sample Bayesian updates. We discretized the grids over μ, $sigma$, y (exemplars) and z (noisy samples) to perform inference, as well as the grid of possible next observations when computing EIG.

To average out noise resulting from grid approximation, we ran each combination of parameter setting and trial type 100 times. To avoid bias, these 100 simulations used 20 different "jitters" - such that the exact location of the points in the discrete grids was randomly chosen. We used the number of samples the model decided to take from each stimulus as a proxy for looking time, effectively converting the binary choice of "look" or "look away" to a continuous variable. We then average the number of samples across the 1000 runs.

This computationally intensive process required significant resources. The grid approximation, combined with repeated simulations, demanded high-performance cluster GPUs to perform vectorized computation using a PyTorch implementation of our model. Running a single parameter setting through an experiment took approximately 10 hours on a GPU - by distributing different parameter settings across GPUs, a full parameter search took about 3 days of compute.

4 RANCH predicts habituation and dishabituation in relation to exposure duration

A key difference between the competing models in Figure 1 is their prediction with regard to the effect of familiarity on looking time. To address this, we collected looking time data from both infants and adults in an experiment where we varied exposure duration to a familiar stimulus, and measured subsequent looking time to a novel vs. familiar stimulus. We then let RANCH "see" the same stimuli and procedure used in our behavioral experiments and evaluated our model's behavior against the behavioral datasets.

4.1 Behavioral experiments

4.1.1 Infant experiment

We tested infants in a novel experimental paradigm in which exposure duration to the stimuli were manipulated in a within-subject design 3 (Experiment 1, Infants). Each infant saw six blocks. In each block there was an familiarizaiton phase and a test trial. During the familiarization, infants were presented with one stimulus, repeated 0 to 9 times. After the familiarization, infants' looking time was measured on a test trial. The test trial showed either the same stimulus as the preceding trials, or a new stimulus. We ran three versions of this experiment, each with three different familiarization durations. We tested a combined sample of 103 7-10 month old infants, with 31 infants in Exp 1A (0, 4 or 8 repetitions), 35 infants in Exp 1B (1, 3 or 9 repetitions) and 37 infants in Exp 1C (2, 4 or 6 repetitions; total n=103, M_{age}=9.38 months, 47 female).

4.1.2 Adult experiment

We ran a self-paced looking time paradigm experiment on Prolific, collecting data from 470 adult participants 3 (Experiment 1, Adults). The stimuli used in the adult experiment are drawn from the same stimuli set used in the infant experiment. During the experiment, participants were asked to watch sequences of animations at their own pace, and they could press a key to proceed to the next trial whenever they wanted to. The time between the onset of the stimulus and the key press was used as a proxy for looking time. Each sequence of animations has two stimuli: the familiar stimulus and the novel stimulus. The familiar stimulus repeated over and over again within a sequence, whereas the novel stimulus only appeared once and it appeared at either the second, the fourth, or the sixth position in the sequence.

4.1.3 Modeling Procedure

The overall modeling strategy is for RANCH to 'see' the experiment the same way as the participants do. RANCH learned from the embeddings derived from the stimuli used in the experiments, and the modeling procedure mirrored the experimental set-up used in the infants experiment and adult experiment respectively (3).

To simulate the infant experiment, the model was first presented with a series of repeating stimuli. For each stimulus, the model took 5 samples each and then moved on to the next stimulus. This phase is similar to the exposure phase infants went through, since during the exposure phase infants had no control over the progression of the stimuli sequence. After the familiarization sequence, the model was presented with the test trial, which either has a novel stimulus or a familiar stimulus (i.e. the repeated stimulus). The number of samples the model took on this test trial was the "looking time" that the model produced.

To simulate the self-paced adult experiment, RANCH makes decisions on how many samples to take on every trial, until the end of the sequence. As in the adult experiment, the sequence has a familiar stimulus that repeats throughout the sequence, and a novel stimulus that appears once. The model therefore predicts looking times to each stimulus in the sequence.

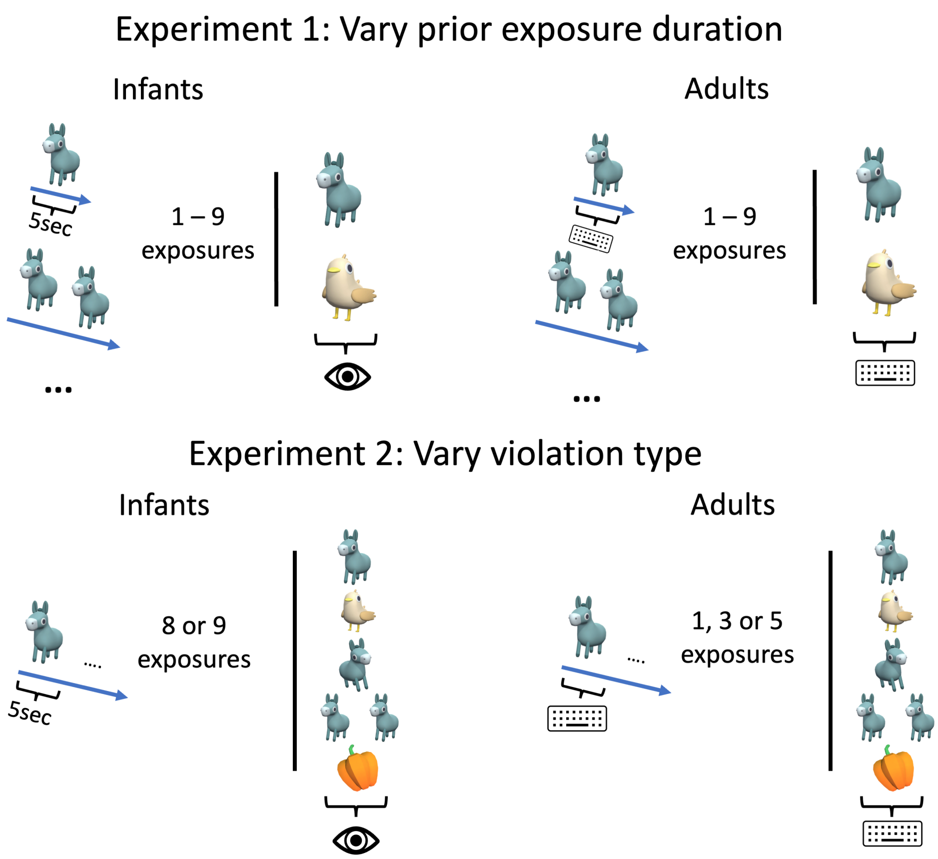

Figure 3: Experimental design

4.2 Evaluating model with behavioral results

The paradigm successfully captured habituation in both infants and adults. In infants, as the number of trials in exposure increased, the looking time at the familiar stimulus in the test trials decreased (β = -4.55; SE = 0.92; t = -4.92; p < 0.001). We also found evidence for infants' dishabituation: looking time for the novel test trials was longer than the familiar trial after long exposures (after 8 repetitions: β = 0.82; SE = 0.22; t = 3.81; p < 0.001, after 9 repetitions: β = 0.51; SE = 0.12; t = 4.21; p < 0.001). Likewise, we also found that adult experimental paradigm successfully captures habituation and dishabituation (Trial number: β = -0.01; Trial Type: β = 0.22; Both SE < 0.001; Both p < 0.001). We found no evidence for familiarity preferences or non-linearities in the shape of the dishabituation curve in infants or adults (cf. 1A&B).

RANCH predicted habituation and dishabituation in both infants and adults. We evaluated each parameter set's performance by running a K-fold crossvalidation test. At each fold, we fit a linear model predicting the looking time data in the fold. We then used the coefficients and the intercepts from the fitted model to scale the model results to the rest of the behavioral data. We compute RMSE and the Pearson's R between the scaled model results and the test set. This process is repeated for each fold and for each parameter setting.

Linking Hypothesis	Infants: Mean RMSE	Infants: Best RMSE
EIG	3.939	3.696
KL	3.912	3.658
Surprisal	4.396	3.601
No learning	4.463	4.135
No noise	4.224	3.875

Table 1: Table 1: Crossvalidation results

4.3 Comparison with alternative linking hypotheses

We also compared the performance of RANCH when using EIG, which is the fully rational linking hypothesis, to other, proxy-based linking hypothesis used in prior literature: Kullback-Leibler divergence (KL-divergence) and surprisal. KL-divergence measures the information gained when one's beliefs are updated, effectively quantifying 'learning progress', while surprisal captures how inconsistent incoming information is with prior expectations. Both have been shown to be associated with infants' looking behaviors [27, 34]. We then compared the model's performance based on the parameter that provides the best fits to the full dataset according to RMSE. Out of the three linking hypotheses, we found that EIG and KL-divergence were comparable to each other (Infants: STATS; Adults EIG: $RMSE = 125.14, I^2 = 0.91$; Adults KL: $RMSE = 129.01, I^2 = 0.9$). But surprisal-based RANCH performed worse in both populations (Infants: STATS; Adults: $RMSE = 205.52, I^2 = 0.84$).

4.4 Comparison with lesioned model

We also examined the importance of RANCH's key assumption. To do so, we created two lesioned models: one model assumes each perceptual sample is noiseless ("No noise model"), and the other model assumes the sampling decision was random instead of based on the learning model ("No learning model"). Both lesioned models provided poor fits for the behavioral datasets (Table **??**). This suggests that the noisy perception assumption and the learning assumptions were both critical to RANCH's performance.

4.5 Parameter Interpretation / Developmental Comparison

One advantage of RANCH is the interpretability of its parameters. Beyond finding that RANCH's fit to the data is robust across parameters, we can also ask which specific parameters maximize that fit. In particular, comparing the best fitting parameters between infants and adults could provide insights into the origin of the developmental differences in behavior (see 'Joint Scaling' under Methods for technical details). When investigating which parameters were most influential in achieving better fit, we found that sd_epsilon played a key role in improving fit. Conceptually, σ_ϵ represents an agent's beliefs about how much noise there is in their own perceptual input. We found that this value tends to be high for infants, and low for adults (Figure @ref(fig:parameter_interpretation)).

4.6 Discussion

Through within-subjects measurements of looking time, we examined the sensitivity of looking to familiar vs. novel stimuli as a function of prior exposure. We did not find evidence for non-linearities predicted by prior models (Fig. 1A & 1B), and instead found that habituation and dishabituation followed monotonic trends as predicted by RANCH (Fig. 1C).

We next turn to test RANCH's ability to make predictions in a novel dataset.

5 RANCH predicts stimulus similarity-based dishabituation in infants and adults

Beyond predicting non-linearities in attentional preferences during encoding (Fig. 1B), Hunter and Ames [25] also predicted that after complete encoding of a stimulus, dishabituation magnitude should be linked to the similarity between two stimuli: the more dissimilar the novel stimulus is to the habituation stimulus, the more participants should dishabituate. We designed an experiment testing the relationship between stimuli similarity and dishabituation magnitude by systematically varying the similarity between the familiar stimuli and the novel stimuli. Instantiating RANCH provides an explicit notion of similarity through the use of ResNet50-derived image embeddings. We can therefore test whether RANCH also predicts such similarity-based differences in dishabituation magnitude. We showed that the best-fitting parameters trained from the previous behaviors predict infants and adults behaviors in the new experiment. In other words, RANCH is capable of making out-of-sample predictions about novel behaviors that it was not trained on.

5.1 Behavioral Experiment

5.1.1 Infant experiment

To study whether infants show similarity-based dishabituation between the habituation and novel stimulus, we ran an experiment via Zoom (n=57, age range=7-10 months, M_{age}=8.9893333months), and an exact replication via Lookit (n =66, age range=6-12 months, M_{age}=9.6853333months). We presented infants with a series of blocks, each again consisting of a familiarization and a test trial (Figure 3, Experiment 2, Infants). Instead of modifying prior exposure as in the previous experiment, here infants were always familiarized 8 or 9 times with an animal (corresponding to the amount of exposure that caused robust dishabituation in the previous experiment). After familiarization, infants saw a test trial, which could relate to the familiar animal in one of five ways: they either saw the same animal again (background), the same animal facing in the opposite direction (pose), a different animal facing the same way (identity), two instances of the same animal side-by-side (number), or an inanimate vegetable or fruit (animacy). In a control experiment, we tested baseline interest in these stimuli without prior familiarization, to ensure that condition differences in the main experiments are due to violations, rather than intrinsic to the stimuli (n=35, age range=7-10 months, M_{age}=9.2096667months).

5.1.2 Adult experiment

This adult experiment was procedurally similar to the previous experiment and has similar adaptation to the infant experiment. Participants (N = 468) saw a sequence of animation, with repeating stimulus and one novel stimulus that appears at the end of the sequence. Since adults have longer attention span, we can test adults in all permutations of the familiar stimulus and novel stimulus. The familiar stimuli were either animals or vegetables, facing eighth left or right, coming in either one or a pair. Like the infant experiment, the novel stimulus can be different from the familiar stimulus in one of the four ways: pose, identity, number, or animacy.

5.1.3 Results and Model Evaluation

We successfully captured similarity-based dishabituation in both infants and adults: Similar to the previous experiment, infants showed dishabituation to all novel test trials (identity: $\beta = 0.33$; $SE = 0.16$; $t = 2.08$; $p = 0.04$, number: $\beta = 0.32$; $SE = 0.17$; $t = 1.92$; $p = 0.06$, animacy: $\beta = 0.33$; $SE = 0.16$; $t = 2.02$; $p = 0.05$), except the pose violation, which elicited no significant dishabituation (pose: $\beta = 0.13$; $SE = 0.16$; $t = 0.83$; $p = 0.41$). There was also evidence that the magnitude of dishabituation was related to the similarity of the novel trial: Infants looking time followed the pre-registered expected ordering: animacy > number > identity > pose > background (Figure REF). A control experiment suggested infants showed no baseline differences in looking time between categories (Figure @ref(fig:control_exp)).

Our exact replication via Lookit showed similar, though not identical results: We again found that animacy and number resulted in the strongest dishabituation in infants (animacy: $\beta = 0.34$; $SE = 0.14$; $t =$; $p = 0.01$, number: $\beta = 0.35$; $SE = 0.13$; $t = 2.68$; $p = 0.01$), and pose violations did not elicit dishabituation ($\beta =$; $SE =$; $t =$; $p =$). However, unlike the Zoom study, the identity violation did not elicit significant dishabituation in our Lookit study ($\beta = 0.07$; $SE = 0.14$; $t = 0.46$; $p = 0.65$). [1]. We suggest potential reasons in the general discussion.

In adults, like in previous experiments, we found evidence for habituation (STATS). More importantly, we also found evidence for graded dishabituation: the dishabituation magnitude to animacy violations is larger than to number violations (STATS) and to pose violations (STATS). Same pattern was found for identity type novelty (STATS). However, animacy type novelty was not different from identity type novelty, nor was the number type novelty different from the pose type novelty (STATS). Qualitatively, we found that the ordering for dishabituation magnitude was animacy (STATS) > identity (STATS) > pose (STATS) > and number (STATS).

We next tested whether RANCH made similar predictions. We used the same modeling procedure described in the previous experiment, simulating the infant and adult experiment respectively. We converted the raw images into the embeddings and assembled the stimuli into the sequences that participants saw in each block.

Using the previously best-fitting parameters in Experiment 1, we found that RANCH generalized to Experiment 2 successfully (Adults: STATS; Infants: STATS). Moreover, RANCH also showed a qualitative ordering of the dishabituation magnitude: animacy > identity > number > pose. This ordering was in line with the CNN-derived embedding distances between the different stimulus categories (Figure @ref(fig:embedding_distances)).

Taken together, though there were some discrepancies between the qualitative ordering between infants, adults and RANCH (discussed below), our analyses show that RANCH is not only capable of capturing habituation and dishabituation, but also can generalize to a new task and make predictions about dishabituation magnitude in relation to stimulus similarity.

6 General Discussion

Measuring habituation and dishabituation has been a key way to gaining insight into the infant mind, but the computational underpinnings are not well understood. Here, we seeked to understand to what extent these

[1] Note that we pre-registered the identity violation as a data quality check, since we had replicated that effect in both infant experiments. However, surprisingly, the identity condition was the only condition that differed from the Zoom experiment, so here we deviate from pre-registration and report these Lookit findings regardless given the high value of having a second sample of infant data.

behaviors can be understood as rational exploration of noisy perceptual samples. To do so, we introduced a rational learner model, RANCH, which takes noisy samples from stimuli and makes sampling decisions based on the next sample's EIG. RANCH makes continuous predictions about infants' looking times to familiar vs. novel stimuli when varying prior exposure. We found that model fit was robust to parameters, and that KL and EIG provided better model fits than surprisal. Furthermore, lesioned models showed that both the learning and noise components of RANCH were essential to its performance. We also found that the key difference between infants and adults was the learner's prior on their noise - the model preferred higher noise to fit the infant data than when fitting the adult data. After fitting RANCH to this exposure duration experiment, we also tested its ability to make predictions on a new task, where we varied the way in which a familiar stimulus is violated by changing its pose, number, identity or animacy. We find that, without any new tuning of the model to the new task, RANCH predicts a gradient in the magnitude of dishabituation similar to the gradient observed in looking time data.

6.1 Novel within-subjects designs

To test our model, we developed novel paradigms for within-infant measurements of habituation and dishabituation. These novel designs allowed us to go beyond testing simple qualitative predictions (i.e. longer versus shorter looking duration). Instead, we validated RANCH's continuous predictions about the relationship between prior exposure and looking to novel vs. familiar stimuli (Experiment 1) and between perceptual distance and looking (Experiment 2), within subjects. Scaling this approach through remote infant behavior testing and automated gaze coding via iCatcher+ [15] has allowed us to collect within-subject data from N infants.

6.2 Graded dishabituation in humans and models

Our findings in Experiment 2 warrants further discussion. In particular, the number violation resulted in interesting patterns across age groups and RANCH: The number of samples RANCH preferred to take from number violations was less than identity violations but more than pose violations, similar to adults. Infants however looked at number violations more than identity violations. This discrepancy between the model and adults on the one hand, and infants on the other hand could be explained by a variety of factors.

RANCH is designed for representing single objects, not multiple objects, therefore any dishabituation to number violations results primarily from simple, low-level perceptual distances, rather than dishabituation to number per-se. To the extent that the dishabituation observed in the data exceeds RANCH's, we can say that this points to other, conceptual sources of surprise that cannot be explained purely by low-level visual differences. This is consistent with infant literature showing that infants are sensitive to changes in number specifically, rather than visual changes generally [40, 17]. Adults on the other hand did not show this exaggerated effect of number violations, possibly because adults' processing was shallower in our task, which is supported by their shorter reaction times.

Infants' stronger dishabituation to number violations than predicted by RANCH was replicated by our Lookit experiment. However, unlike in the original Zoom experiment, the Lookit experiment did not elicit significant dishabituation to identity violations. This is in contrast to our findings in both Experiment 1 and Experiment 2, where identity violations after long familiarizations caused longer looking times than looking to the background. There may be two possible reasons for this discrepancy: First, while in the Zoom study experimenters terminated trials when infants looked away for three seconds, the Lookit experiment used fixed trial durations due to the asynchronous set-up. While in both experiments looking times are still calculated using a standard criterion off-look [24], the infant-contingency of the Zoom experiment likely increases the data quality and results in higher sensitivity to smaller effects. Similarly, an experimenter being present via Zoom allows them to ensure a quiet testing environment and adjust the flow of the experiment to the participant - previous comparisons between Zoom and asynchronous studies similarly suggest that effects via Zoom tend to be larger.

6.3 Unique strengths of RANCH framework

A key strength of RANCH is the cognitive interpretability of its parameters. Previous Bayesian models of infant attention have leveraged parameter interpretability to draw conclusions about the computational mechanisms underlying individual differences in attention [36]. Here, we use parameter fits of RANCH to study population differences: By applying a "joint scaling" procedure in which we scale model samples to infant and adult simultaneously, we could identify which parameters had to be different between infant and adults to account for their different patterns of looking time. When searching for parameters, we found that a key developmental difference was that learner's beliefs about the magnitude of their own noise is higher in infants than for adults.

Another strength is RANCH's ability to model looking on a more fine-grained timescale. Prior research has largely resorted modeling on the trial-level, and asking about correlations between behavior on a given trial (e.g. looking time) and certain information-theoretic proxies, such as KL or surprisal [27, 35]. However, a proper rational analysis with EIG is not possible at this resolution, because models that carve up time in terms of trials, rather than samples, do not allow for asking questions about future information gain. It is only in moment-by-moment modeling frameworks like RANCH that we can implement the formal rational analysis [31, 1], in which learners ask "How much information do I expect to gain from taking another sample from the stimulus?". The ability to conduct the rational analysis enhances our understanding of habituation as optimally adapted to environmental constraints [see e.g. 18, for a different rational analysis of habituation]. Still, replacing EIG with information-theoretic proxies in our setting can help us examine whether participants' behavior seems to follow an EIG-based, or a proxy-based, linking hypothesis. In our analyses we found that though KL divergence performed comparably to EIG, the surprisal-based model was worse at fitting both infant and adult data.

6.4 No evidence for familiarity preferences

A key feature of our experimental data, as well as our model, is that at no point, even for the shortest familiarization durations, we find longer looking to familiar than to novel stimuli. This finding contrasts with predictions made by a prominent model of infant attention (Figure 1A)), which posit that at intermediate levels of encoding, familiar stimuli should become more interesting than novel stimuli. The intuition is that some initial familiarity with a stimulus may allow a viewer to "break into" the information there is to gain from the stimulus - a completely novel stimulus may be too complex or foreign for extended attention to be worth it. In the learning setting of simple perceptual learning, the rational analysis and participants do not seem to favor this strategy. This suggests that limited exposure does not generically lead to familiarity preferences in any experimental paradigm.

Of course, the absence of familiarity preferences in our results does not rule out their existence in general. First, familiarity preferences may be more subtle than novelty preferences, so that the statistical power that is needed to find familiarity preferences is higher than that achieved in the current study. A current large-scale study by the ManyBabies consortium which aims to test the predictions made by Hunter and Ames [25] may give insight into this possibility [28]. In addition, the learning problem that participants are solving likely plays a critical role in whether they will exhibit familiarity preferences. This context-dependence is reflected in meta-analyses investigating familiarity preferences in different paradigms. For example, when tested on word segmentation in their native language, infants show persistent preferences for familiar stimuli throughout the first year [5]. In contrast, when tested on statistical learning of novel words, infants show consistent preferences for novel stimuli, from 4-month- to 11 months of age [6].

6.5 Formal theories of learning and attention

These seemingly contradictory results highlight the need for theories that formalize accounts of how the learning problem influences attention. Dubey and Griffiths [14] give an example of such a formal account by showing that when past and present events are correlated, rational agents, under some assumptions, develop a tendency to attend to familiar stimuli to prepare for the most likely future events, while in uncorrelated environments, novelty preferences are optimal. On the other hand, ideal learners attempting to maximize their expected information gain consistently seek novelty when trying to learn a single concept. Formal models of the learning problem being solved in experiments may therefore give more principled and precise predictions beyond identifying factors like "prior exposure" as determinants of attentional preferences.

Importantly, RANCH's predictions outlined in this paper apply to the specific case: the learner makes noisy observations of 1) stimuli from a specific embedding space, 2) updates its beliefs about the location of a single concept and 3) makes sampling decisions based on EIG. The modular structure of RANCH makes it so that each of these components (embedding space, learning model and linking hypothesis) could be modified, likely leading to different predictions. For example, the embedding space could be chosen to more closely resemble a psychologically validated perceptual space [e.g. 29], though in this case, we found that such an alternative did not significantly impact model fit (REF to SI with Lee & DiCarlo model). Similarly, the learning problem could be modified to account for the idea that participants may not just be learning a single concept, but rather are also deciding whether a stimulus should prompt the creation of a new concept or not.

6.6 Conclusion

Infants are active learners. Since birth, they actively explore their environment through selective attention. Looking time, a widely used measure in developmental psychology, leverages this ability to help make inferences about infants' conceptual and perceptual capacity. However, past literature tends to use looking time measure as a qualitative outcome (e.g. infants either look longer or shorter at a particular stimuli), rather than a graded, continuous measure (e.g. how long infants would look at certain stimuli). In our work, we provided strong evidence that looking time is a graded measure and can be predicted quantitatively by a model that posits that infants' looking time is driven by rational curiosity. This work not only puts the methodological foundations of infant looking time research on firmer ground, but also contributes to our understanding of rational exploration behaviors throughout developmental trajectories.

7 Methods and Materials

7.1 Model Specifications

7.1.1 Perceptual representations

We derived principled low dimensional representations of our stimuli from ResNet50 [22]. We ran a Principal Component Analysis and used the first three principal components. This decision was made due to the computational demands of our model. Each added dimension would increase the total run time exponentially. The first three components capture 57.9% of the variance.

Given that we were using discrete grid approximation in our inference, the model was sensitive to the absolute values of the embeddings. When these embeddings values were close to or exceeded the range of the approximation grids, this would cause bias in our simulation result. We therefore scaled our embeddings down to fall squarely into our grid. This downscaling had to be more extreme for the stimulus type experiment, where embedding distances were larger, in particular for animacy violations. We ran simulations to confirm that embeddings that were too large would result in biased model behaviors (Figure @ref(fig:wrong_scaling)).

7.1.2 Linking Hypotheses

We compared two additional information theoretic measures with Expected Information Gain (EIG): Kullback–Leibler divergence and Surprisal. Each of these measures has been invoked in the literature as a plausible account for explaining looking behaviors across developmental stages [KL divergence: 34, Surprisal: 27].

Our key linking hypothesis, EIG, is a forward-looking measure. It is computed as the product of the posterior predictive probability of the next perceptual sample ($p(z_{t+1}|\theta_t)$) and the information gained conditioned on that next hypothetical sample, via a grid approximation of possible subsequent samples (S).

Surprisal is calculated as the negative log likelihood of the current perceptual sample under the prior distribution

We ran individual simulation and parameter search for each of the three linking hypotheses to ensure fair comparison.

7.1.3 Scaling Procedure for Developmental Interpretation

To find the best fitting parameters for each age group we employ a strategy we call “joint scaling”. As in Fig. **FIXME**, to evaluate model fit, we first map model samples onto looking times by finding the best linear scaling of model samples in a regression of the form `LT ~ model_samples`. The parameter set that minimizes the RMSE between the scaled model samples and looking times is what we consider the best parameter set. In the case of “joint scaling”, we regress both the adult and the infant simulations against their looking times simultaneously, to find how varying the infant and adult parameters affect the fit of the scaled model samples to the adult and infant data. This joint scaling procedure puts pressure on the parameters, rather than the scaling, to account for the fact that adults’ looking times are a lot shorter than infants’ looking times.

The interpretable parameters of RANCH are the priors on μ and σ and the priors on the noise. The priors on μ and σ were parameterized by a normal inverse-gamma prior with μ_{prior}, ν_{prior}, α_{prior} and β_{prior}, the conjugate prior to a normal distribution with unknown mean and variance. While the mean was fixed to be 0 (μ_{prior}), variation in the other parameters express distinct hypotheses about the precision of the location of the concept (ν_{prior}), as well as the variance of the concept (α_{prior} and β_{prior}). The learner’s prior on noise are parameterized by gaussian distribution with a mean (μ_{noise}) and standard deviation (σ_{noise}). In our parameter search, we only searched over the σ_{noise}.

7.2 Experimental Procedure

Below we describe in detail the experimental procedure. We use Experiment 1 to denote the experiment that manipulates exposure duration, and Experiment 2 to denote the experiment that manipulates stimuli similarity.

7.2.1 Infant experiment

7.2.1.1 Participants For Experiment 1, we tested a combined sample of 103 7-10 month old infants, with 31 in Exp A, 35 in Exp B and 37 in Exp C (103 (M_{age} = 9.38 months, 47 female). 10 participants were excluded completely due to fussiness. An additional 58 individual test events were excluded (out of 552.

For Experiment 2, we tested a combined sample of 123 7-10 month old infants, with 57 in the Zoom experiment (7 complete exclusions, 42 trial exclusions), 66 in the Zoom experiment (3 complete exclusions, 26 trial exclusions) and 35 in the control experiment (2 complete exclusions, 10 trial exclusions).

Complete or trial exclusions occurred because 1) infants fussed out of the experiment at an earlier stage of the experiment, 2) infants looked at the stimuli for less than a total 2 seconds, 3) there were momentary external distractions in the home of the infant or 4) the gaze classifier performed so poorly that manual corrections were not viable.

7.2.1.2 Stimuli

In Experiment 1, we presented infants with a series of animated animals, created using "Quirky Animals" assets from Unity (Figure 3), link to assets). The animals were walking, crawling or swimming, depending on the species. In a previous set studies, not included here (but see OSF repository), we showed infants a different stimulus set and failed to elicit replicable habituation, novelty or familiarity preferences. As a result, we modified the stimuli to be more appealing, and separated familiarization trials with curtains opening and closing to accentuate familiarization trials as separated occurrences of the stimulus.

In Experiment 2, (Figure 2A, we introduced animacy violations using the "3D Prop Vegetables and Fruits" link to assets) assets from Unity and animated bouncing and rotating trajectories for them. To create number violations, we duplicated the videos of single animal Unity assets and pasted them side-by-side.

7.2.1.3 Procedure

Data collection was performed primarily synchronously on Zoom, except for the Lookit replication of Experiment 2. In all cases, infants were recruited from Lookit [**scott2017lookit**] and Facebook. Parents were instructed to find a quiet room with minimal distractions, place their child in a high chair (preferred) or on their lap, and to remain neutral throughout the experiment. Infants were placed as close as possible to the screen without allowing them to interact with the keyboard.

Both main experiments followed a block structure, where each block was divided into two sections: 1) a familiarization period and 2) a test event. Each block was preceded by an "attention getter", a salient rotating star. During the familiarization period, the infant was familiarized to a particular animal, the background, in a series of familiarization trials. Each familiarization trial was a 5 second sequence: curtains open for 1 second, the animated animal moves in place for 3 seconds, and then the curtains close for 1 second. In the exposure duration experiments, the number of familiarization trials (the "exposure duration") varied between blocks. In the stimulus type experiment, the number of familiarizations was always 8 or 9.

During the test event, the infant saw either the same background animal again, or a novel stimulus, the deviant. The onset of the test event was not marked by any visual markers, but a bell sound played as the curtains opened, to maximize the chance of engagement during the test event. On Zoom, the test event used an infant-controlled procedure: the experimenter terminated the trial when the infant looked away for more than three consecutive seconds. On Lookit, test trials would always be played for 40 seconds. In both cases, looking time was then defined as the total time that the infant spent looking at the screen from the onset of the stimulus until the first two consecutive seconds of the infant looking away from the screen. If the infant did not look away after 60 seconds of being presented with the test event, the next block automatically began and infants' looking time for that test event was recorded as 60 seconds.

In Experiment 1, the deviant stimulus was always another animal. In Experiment 2, one of four properties of the background animal could be violated: pose, identity, number or animacy.

In Experiment 1, each infant saw six blocks: Three different exposure durations (0, 4 and 8 in Exp. A, 1, 3 and 9 in Exp. B and 2, 4 and 6 in Exp. C) appeared twice each, once for each test event type (background or deviant). The longer exposure durations (8 or 9) were chosen based on our previous pilot studies with a different stimulus set (OSF repository), and the shorter durations were chosen to provide limited learning experience with the background. The order of blocks was counterbalanced between infants, and pairs of animals (background and deviant) were counterbalanced to be associated with each block type. Which animal was shown as the background and which as the deviant (if it was a deviant test trial) was randomized in each block.

In Experiment 2, each infant saw four blocks within a session - the background trial was always one of the four test trials, and the remaining three test trials were picked randomly from the remaining four violation types (pose, identity, number, animacy). The control experiment showed six standalone trials: Infants saw two single animals, two pairs of animals and two single vegetables, to measure baseline interest in these stimulus categories.

7.2.1.4 Looking time coding

To code the infants' looking time we used iCatcher+, a validated tool developed for robust and automatic annotation of infants' gaze direction from video [**erel2022icatcher**]. To obtain trial-wise looking time, we merged iCatcher+ annotations with trial timing information, thereby fully replacing manual coding of looking time. The correctness of iCatcher+ coding was manually supervised to check for failed face detections or distractions.

7.2.1.5 Statistical analysis

We ran the pre-registered statistical analyses for both experiments. All statistical analyses were ran in R using the package `lme4` [4].

For Experiment 1, we ran a `lme4` model with the following specification: `log(looking time) ~ poly(exposure duration, 2) * test type [identity or background] + log(block number) + (1|subject)`. The `exposure duration` term refers to the amount of prior exposure received by the infant (0 to 9 prior exposures), and the second polynomial tested for non-linearity in the effect on looking time, as predicted by some theories of infant attention. This non-linearity could occur primarily for looking to the background (Figure 1A), or both background and novel stimuli (Figure 1B), hence the interaction with `test type`. We control for `block number` to account for general decreasing interest in our stimuli as the experiment goes on.

For Experiment 2, we ran a linear mixed effects regression, again predicting looking time, using violation type as the main predictor: `log(looking time) ~ test type [animacy, pose, identity, number, background] + log(block number) + (1|subject)`. Here, we set `background` as the reference level for `test type`, and asked whether the other test types were significantly different from that background reference level. We again controlled for block number.

7.2.2 Adult experiment

7.2.2.1 Participants

Both experiments were run on Prolific (Experiment 1: N = XXX; Experiment: N = XXX). Two experiments have the same pre-registered exclusion criteria. We excluded participants if (1) the standard deviation of their reaction time across all trials was less than 0.15 (indicating key-smashing, Experiment 1: N = XXX; Experiment 2: N = XXX), (2) spent more than three absolute deviations above the median of the task completion time as reported by Prolific (Experiment 1: N = XXX; Experiment 2: N = XXX), and (3) provided the wrong response to more than 20% of the memory task (Experiment 1: N = XXX; Experiment 2: N = XXX). A total of XXX and XXX participants met at least one of the three criteria and were excluded from the final analysis for Experiment 1 and Experiment 2, respectively. We also excluded a trial if the trial was three absolute deviations away from the median in the log-transformed space across reaction time from the remaining participants.

The final sample included XXXX participants (Experiment 1: Mean Age, SD Age; Experiment 2: Mean Age, SD Age).

7.2.2.2 Stimuli

The stimuli used in adult experiment are identical to the ones used in infant experiment. In Experiment 1, the stimuli are selected from the animal unity set. In Experiment 2, both the animal and vegetable sets were used.

7.2.2.3 Procedure

Procedurally, Experiment 1 and Experiment 2 are similar. Both are self-paced visual presentation studies. Participants were instructed that they would be looking at a sequence of animated stimuli at their own pace. At each trial, then can press a key on the keyboard to move on to the next trial after a minimum viewing time of 500 ms. Each study consists of multiple blocks. Each block consists of multiple trials. Between blocks, participants answered a simple memory question ("Have you seen this animation before?"). This memory question is used as an attention check.

Each block contains multiple trials of one stimulus being repeatedly presented (the familiar trials) and one trial that shows a different stimulus (the novel trial). The novel trial always appears as the last trial. In Experiment 1, the total number of trials range from two to eleven. The novel trial was randomly drawn from the stimuli set that researchers haven't seen before. In Experiment 2, the block consists of either two, four, or six trials. There are a total of eight types of stimuli in the familiar trials. They are all combinations of the three features that each include two levels: animacy (e.g. animate or inanimate), number (singleton or pair), and pose (facing left or facing right). The novel trial can differ from the familiar trials in one of the four dimensions: pose, number, identity, and animacy. Both experiments also have blocks that only include one stimulus (the familiar block).

The order of the block is semi-randomized in both experiments. Experiment 1 has 15 blocks in total. We grouped the total number of trials in each block into five pairs of two: [2, 3], [4, 5], [6, 7], [8, 9], [10, 11], and randomly sampled 1 block length from each pair. In other words, each participant would receive 5 different lengths of repeating background trials, stratified by 5 levels. We controlled the distribution of the block lengths by making sure the first half and the second half of the experiment each has the same number of blocks with different block lengths. Experiment 2 had 24 blocks in total. We grouped the twenty-four blocks into four groups. Each group consisted of two familiar blocks and one block from each of the four novelty types. The order of blocks within each group was randomized.

7.2.2.4 Statistical Analysis We ran the pre-registered statistical analyses for both experiments. All statistical analyses were ran in R using the package `lme4` [4].

For experiment 1, we ran a linear mixed effect regression predicting the log transformed of looking time with the following specification: `log(looking time) ~ trial_number + is_first_trial + trial_number * trial_type + log(block_number) + (trial_number * trial_type | subject) + (is_first_trial + trial_number | subject)`. This model allows us to investigate whether the number of familiar trials has an impact on the looking time (i.e habituation), and whether the trial type (familiar or novel) mediates the influences. This model failed to converge. Following the pre registered procedure, we pruned the model to include only by-subject random intercept. The pruned model suggests that there is evidence for habituation and dishabituation (STATS).

For experiment 2, we ran a similar model and incorporated the effect of different novelty type. The particular model specification is as follows: `log(total_rt)` $\sim$ `trial_number + is_first_trial + (trial_number + is_first_trial) * stimulus_number + (trial_number + is_first_trial) * stimulus_pose +(trial_number + is_first_trial) * stimulus_animacy + (trial_number + is_first_trial) * novelty_type + log(block_number)`. The `novelty_type` has five levels, including the background trial and four types of violation.This model can help us test two hypotheses: (1) whether our experimental paradigm captured habituation and dishabituation and (2) whether the magnitude of dishabituation was influenced by the similarity between the novel trials and the familiar trials.

8 Supplementary Information

8.1 Parameter robustness

Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness.Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness. Some text about parameter robustness.

8.2 Scaling the embedding space

In order to achieve sensible results, it was important to bring the stimulus embeddings in line with the approximation grid used for inference. Below we show what happens embeddings exceeded or came close to

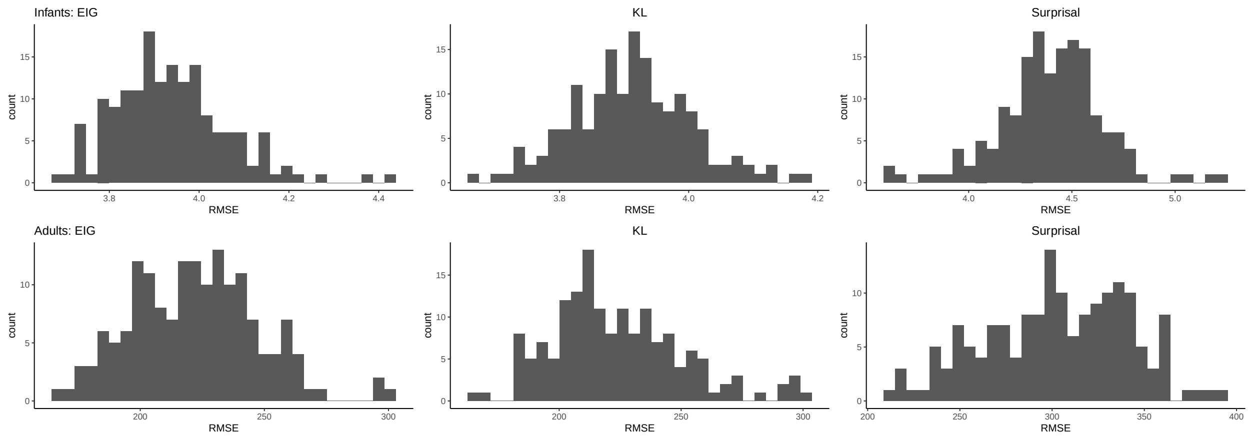

Figure 4: (#fig:parameter_robustness)Histograms of RMSE for different linking hypotheses show that RANCH is largely robust to variations in parameters.

the edges of the our approximation grids. In Experiment 1, unscaled embeddings resulted in a reversal of background vs. deviant model samples, such that the model takes more samples from the background than the deviant. In Experiment 2, unscaled embeddings resulted in an unintuitive ordering which is not in line with the embedding distances (Figure @ref(fig:embedding_distances)).

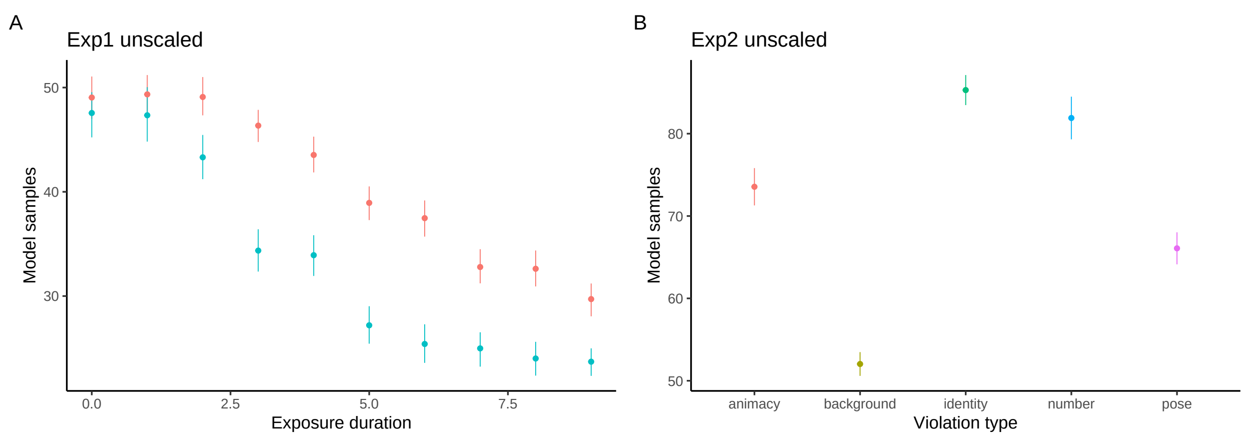

Figure 5: (#fig:wrong_scaling)The effect of scaling on model results. A) Experiment 1 with unscaled embeddings results in a reversal of background vs. deviant. B) Experiment 2 with unscaled embeddings results in an unintuitive ordering.

8.3 Developmental comparison of parameter fits

A naive approach to comparison parameter fits between infants and adults would be the following: After joint scaling, pick the set of parameters for adults and infants that together minimize RMSE. If we do this, we obtain the following parameter sets:

infants: ν= 1, α= 1, β= 0.1, σ_ϵ= 1, σ_ϵ= 1, EIG_{env}= 10^{-5}

adults: ν= 3, α= 10, β= 0.1, σ_ϵ= 0.1, σ_ϵ= 0.1, EIG_{env}= 10^{-4}

If we compare these two sets of parameters to each other, and interpret these differences, we may come to the following conclusion: RANCH fits best when infants' σ_ϵ is higher than that of adults, and β_prior is higher than that of adults, suggesting that infants both expect their perception to have more noise, and that infants have more prior uncertainty about the size of the standard deviation.

However, a direct comparison between these parameter sets does not give insight into the relative stability of RANCH's fit to the data if we change any of these parameters: It may be that if a parameter was slightly different, it would not make a big difference for model fit.

In order to investigate this, we want to also check the sensitivity of the model fit to changes in the parameters. To do so, we plot RMSE histograms colored by the value of a certain parameter, from a certain age group. Here we see that the overall modes of the RMSE histograms are dominated by `world EIG`, likely because this parameter is the main threshold that can account for the difference in scale between infant and adult looking times. However, of the other parameters, we see that RMSE only shows sensitivity to σ_ϵ. Other parameters that are different between the best fits do not show this sensitivity. We therefore conclude that the main interpretable parameter difference is in σ_ϵ, the learner's noise prior, being higher in infants than in adults.

Figure 6: (#fig:parameter_interpretation)Histograms of model fit colored by priors. Results suggest that when the learner's noise is high, RANCH fits infant data better.

8.4 Embedding distances of different violations categories, by embedding model

Below we plot the embedding distances of the different violations. The lines connect a single stimulus, and the y-axis represents the embedding distances to all the stimuli that would constitute a certain violations. For example, for a single animal stimulus, the y-value for 'animacy' corresponds to the distance between the single animal and all the single fruit/vegetable stimuli.

8.5 Control experiment for Experiment 2

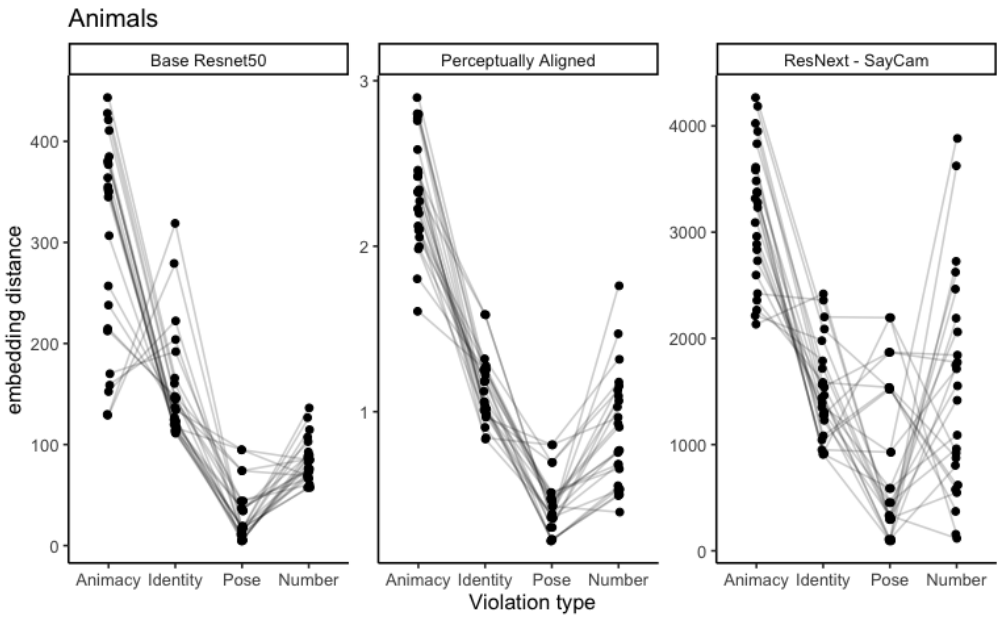

Figure 7: (#fig:embedding_distances)Embedding distances of different violation types for 1) a base ResNet50 model, 2) a perceptually aligned embedding model (Lee, 2022) and 3) a ResNet50 model trained on SAYCAM data (Orhan et al., 2020)

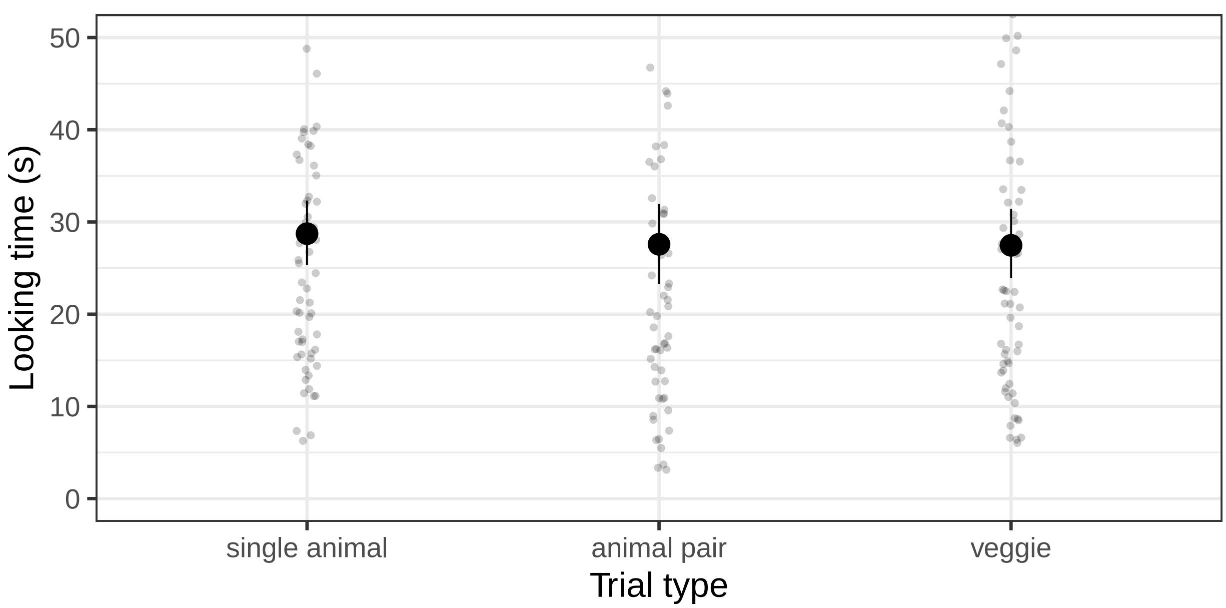

Figure 8: (#fig:control_exp)Control experiment for experiment 2, checking for baseline differences in interest.

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